

FULL STACK DEVELOPMENT WITH NODE JS



HTML 5

- Introduction HTML
- HTML Basics
- HTML Elements
- HTML5 Semantic
- HTML Attributes
- HTML Headings
- HTML Paragraph
- HTML Styles
- HTML Formatting
- HTML Quotations
- HTML Computer Code
- HTML Comments & Colors
- HTML CSS, Links and Images
- HTML Lists
- HTML Blocks
- HTML Classes
- HTML Layout
- HTML Responsive
- HTML iframes
- HTML JavaScript
- HTML Head
- HTML Entities and URI Code
- HTML Symbols and XHTML
- HTML Charset and Forms
- Quiz
- Assignment

Css3

- Introduction CSS3
- CSS3 Syntax
- CSS3 How To
- CSS3 Colors
- CSS3 Backgrounds
- CSS3 Borders
- CSS Padding
- CSS Height/Width
- CSS3 Gradients
- CSS3 Shadows
- CSS3 Text
- CSS3 Fonts
- CSS3 2D Transforms
- CSS3 3D Transforms
- CSS Links
- CSS Lists
- CSS Tables
- CSS Box Model
- CSS Outline
- CSS Display
- CSS Max-width
- CSS Position
- CSS Float

- CSS Inline-block
- CSS Align
- CSS Combinators
- CSS3 Media Queries
- CSS3 Responsive
- Quiz
- Assignment

JavaScript

- Introduction to JavaScript
- Java Script Language Basics
- JavaScript Objects
- JavaScript Scope
- JavaScript Events
- JavaScript Strings
- JavaScript Numbers
- JavaScript Math
- JavaScript Arrays
- JavaScript Boolean
- JavaScript Comparisons
- JavaScript Conditions
- JavaScript Switch
- JavaScript Loops
- JavaScript Type Conversion
- JavaScript RegExp
- JavaScript Errors
- JavaScript Debugging
- JavaScript Hoisting
- JavaScript Strict Mode
- JavaScript Functions
- JavaScript Objects
- JavaScript Forms
- JavaScript HTML DOM
- JavaScript BOM
- Quiz
- Assignment

jQuery

- Introduction to jQuery
- jQuery Syntax
- jQuery Selectors
- jQuery Events
- jQuery Effects
- jQuery HTML
- jQuery Traversing
- jQuery AJAX & Misc.
- Quiz
- Assignment

Bootstrap 4

- Introduction to Bootstrap
- Bootstrap Basics
- Bootstrap Grids
- Bootstrap Themes
- Bootstrap CSS
- Bootstrap JS
- Quiz
- Assignment

Advance JavaScript (ES 6)

- Introduction to ES 6
- Let and Constant · Arrow Function
- Template Literals
- Class and Inheritance
- Rest parameter and Spread Operator
- Map ,Filter and Reduce function
- Modules
- Destructuring
- For in and For of loop

Typescript

- Introduction to Typescript
- Environmental Setup· Types
- Object Oriented Features
- Interfaces
- Modules

INTRODUCTION TO REACT JS

- What is React JS?
- What is SPA?
- DOM vs Virtual DOM
- Advantages and Disadvantages
- Key Features

ENVIRONMENTAL SETUP

- Node | NPM
- Installation of CLI
- Setup Project
- Directory Structure
- Code Editors
- How React JS Application Boot

BASIC FEATURES OF REACT JS

- React Concepts
- JSX and TSX
- Render Elements
- Function and Class Components
- Props and State
- Handling Events
- Dynamic Data Rendering
- Property Binding

KEY FEATURES OF REACT JS

- Conditional Rendering
- List and Keys
- Forms Handling
- Forms Validations

COMPONENT LIFECYCLE HOOK

- Understanding component lifecycle
- All Lifecycle Hooks

EVENT HANDLING REACT

- Understanding React Event System
- Passing arguments to event Handlers

NETWORK CALL

- Fetch
- Axios

CUSTOM SERVICES

- Introduction to Services
- Building a Service

LOCAL DATA STORAGE

- Local Storage
- Session Storage
- Cookies

ROUTING WITH REACT ROUTER

- Setting up React Router
- Configuring route with Route Component
- Making routes dynamic with Route Params
- Working with nested routes
- Link and NavLink
- Redirect Routes

UI COMPONENTS

- Angular Material
- PrimeNG

INTRODUCTION TO REDUX

- Why Redux
- Install and setup
- Store ,Reducer , actions
- Dispatcher
- High order Components
- mapStateToProps and mapDispatchToProps usage

ADVANCE REDUX

- Async Actions
- Middleware
- Redux Thunk and Redux Saga

React Hooks

- Why We Need Hooks.
- Different Types Of Hooks
- Using State And Effect Hooks
- Usereducer , Useref Etc.
- Custom Hooks
- Rules Of Hooks

Third Party Modules

- Social Login
- Pagination
- Search
- Filter
- JWT Token
- File Upload
- Many More

Rest Js Testing

- Jest with Enzyme

Next js

- Introduction
- Routing
- Page Pre-Rendering (Server Side Rendering) & Data Fetching
- Server-side rendering vs Client-side rendering
- Optimizing NextJS Apps
- Adding Backend Code with API Routes
- Working with App-wide State
- Adding Authentication

Ecommerce Project

Node Js

Introduction to Node.js

- What is node.js
- A word on event loop
- Event driven architecture
- Blocking vs non-blocking code
- Node Process Model
- Setup node.js
- Node REPL
- Debugging node.js app

Modules in JS

- Understanding exports and require
- Creating modules
- Importing modules
- Quick tour on NPM
- Installing 3rd party modules

Events & streams

- Significance of Events
- Event Emitter class
- Emitting and listening to events
- Types of streams
- Working with streams
- Composing streams using pipe

Working with Process and File System

- Process object
- Handling exceptions at process level
- Listening to process events
- Working with file system using fs module
- Creating, Copying, Deleting files
- Creating, Copying, Deleting directories
- Watching for file changes

HTTP

- Building a Web Server
- HTTP Request Methods, Headers
- Response Codes and Headers

Express Js

Expressjs Introduction & Setup

- Building a Web Server
- HTTP Request Methods, Headers
- Response Codes and Headers
- Serving static files with express
- Express application generator

Routing

- Route Methods
- Route Paths
- Parameterized Routes
- Route Handlers
- Express Router

Middleware

- Types of middleware
 - Application
 - Router
 - Error
 - Built-in
 - Third-party
- Error middleware
- Body parser, cors , cookie parser, session management

Template Engines

- What are template engines
- EJS
- Jade
- Handlebars

Expressjs security

- Authentication
 - JWT
 - Securing routes
- Debugging in Express
- Realtime programming with Socket.io

Expressjs security

- Authentication
 - JWT
 - Securing routes
- Debugging in Express
- Realtime programming with Socket.io

MongoDB

Introduction to MongoDB

- Installing MongoDB
- The current SQL/NoSQL landscape
- Document-oriented vs. other types of storage
- Mongo's featureset
- Common use-cases
- MongoDB databases
- MongoDB Collections
- MongoDB Documents

CRUD Operations in MongoDB

- Creating documents
 - insert()
 - update()
 - save()
- Querying documents
 - find()
 - Working with equality
 - Query operators
 - Building complex queries
- Updating documents
- Deleting documents
- Limit
- Order By

Introduction to Mongoose

- Word on ORM/ODM
- Installing mongoose
- Connecting to MongoDB from mongoose

Core concepts of Mongoose

- Understanding mongoose schemas and datatypes
- Working with Models
- Using modifiers in schema
- Using virtual fields
- Optimizing query performance by enabling indexes

FullStack Application Deployment

- Git
- Build Application
- Deploy in Heroku
- Deploy in AWS



Partners :



Java



E-mail: info@ducatindia.com
Visit us: www.ducatinidna.com
www.facebook.com/ducateducation

NOIDA SEC-16

A-43 & A-52, Sector-16,
Noida - 201301, (U.P.) INDIA
☎ 70-70-90-50-90
📠 +91 99-9999-3213

2.0 NOIDA SEC-63

H-43 Sector-63
Noida-201301
☎ 70-70-90-50-90
📠 +91 7042175774

3.0 NOIDA EXTENSION

40 A, TechZone 4, Near Ashoka
Tower, Greater Noida West,
Uttar Pradesh - 201009
☎ 72-89-88-66-55
📠 +91-8448160652

PITAMPURA (DELHI)

Plot No. 366, 2nd Floor,
Kohat Enclave, Pitampura,
(Near- Kohat Metro Station)
Above Allahabad Bank,
New Delhi- 110034.
☎ 70-70-90-50-90

SOUTH EX (DELHI)

D-27, South Extension-1
New Delhi-110049
☎ 70-70-90-50-90
📠 +91 98-1161-2707

VIKASPURI (DELHI)

4, A-Block, Vikaspuri, Delhi,
Nearest Metro Janakpuri West,
Pillar No. 628, New Delhi - 110018
☎ 85-95-52-67-42
📠 +91 98-1161-2707

GURGAON

1808/2, 2nd floor old DLF,
Near Honda Showroom,
Sec.-14, Gurgaon (Haryana)
☎ 70-70-90-50-90

FARIDABAD

16/5 Mathura Road, Sec 19,
Behind Metro Pillar No 687,
Faridabad
☎ 70-70-90-50-90
📠 +91-9355333011

JAIPUR

55 Shrigopal Nagar,
Gopalpura Bypass, Near
Mahesh Nagar Police
Station, Jaipur - 302015
📠 +91-9090708058